

Brendan Boyd

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biboyd.github.io

Education

SUNY Stony Brook University Physics & Astronomy Department <i>PhD in Physics, with Concentration in Astronomy</i>	2020-2026
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Michigan State University College of Natural Science Honors College <i>Bachelor of Science, Astrophysics</i> <i>Minors in Math and CMSE</i>	2016-2020
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Publications

Refereed Journal Articles

Brendan Boyd et al. “Simulating the Convective Urca Process with Multiple Urca Pairs in a Simmering White Dwarf”. In: *arXiv e-prints*, arXiv:2605.19968 (May 2026), arXiv:2605.19968. DOI: 10.48550/arXiv.2605.19968. arXiv: 2605.19968 [astro-ph.SR]

Ferran Poca-Amorós et al. “On the Importance of the Convective Urca Process in 3D Simulations of a Simmering White Dwarf”. In: *The Astrophysical Journal* 1000.2 (Mar. 2026), p. 298. DOI: 10.3847/1538-4357/ae4a9b. URL: <https://doi.org/10.3847/1538-4357/ae4a9b>

Brendan Boyd et al. “3D Convective Urca Process in a Simmering White Dwarf”. In: *The Astrophysical Journal* 979.2 (Jan. 2025), p. 216. DOI: 10.3847/1538-4357/ad9bb0. URL: <https://dx.doi.org/10.3847/1538-4357/ad9bb0>

Alexander I. Smith et al. “pynucastro: A Python Library for Nuclear Astrophysics”. In: *The Astrophysical Journal* 947.2, 65 (Apr. 2023), p. 65. DOI: 10.3847/1538-4357/acbaff

Refereed Conference Proceedings

B. Boyd et al. “Approximating Convective Urca Cooling in a Simmering White Dwarf”. In: *Journal of Physics Conference Series*. Vol. 2997. Journal of Physics Conference Series. IOP, Apr. 2025, 012006, p. 012006. DOI: 10.1088/1742-6596/2997/1/012006

B. Boyd et al. “Sensitivity of 3D Convective Urca Simulations to Changes in Urca Reactions”. In: *Journal of Physics Conference Series*. Vol. 2742. Journal of Physics Conference Series. IOP, Apr. 2024, 012001, p. 012001. DOI: 10.1088/1742-6596/2742/1/012001. arXiv: 2311.07743 [astro-ph.SR]

Zhi Chen et al. “A Framework for Exploring Nuclear Physics Sensitivity in Numerical Simulations”. In: *Journal of Physics Conference Series*. Vol. 2742. Journal of Physics Conference Series. IOP, Apr. 2024, 012021, p. 012021. DOI: 10.1088/1742-6596/2742/1/012021

Alexander Smith Clark et al. “pynucastro 2.1: an update on the development of a python library for nuclear astrophysics”. In: *Journal of Physics Conference Series*. Vol. 2742. Journal of Physics Conference Series. IOP, Apr. 2024, 012003, p. 012003. DOI: 10.1088/1742-6596/2742/1/012003

Software

Ann Almgren et al. *AMReX-Astro/MAESTROeX: MAESTROeX 25.02*. Version 25.02. Feb. 2025. DOI: 10.5281/zenodo.14811053. URL: <https://doi.org/10.5281/zenodo.14811053>

AMReX-Astro initial_models team et al. *AMReX-Astro/initial_models: Release 24.03*. Version 24.03. Mar. 2024. DOI: 10.5281/zenodo.10827428. URL: <https://doi.org/10.5281/zenodo.10827428>

Brendan Boyd et al. "SALSA: A Python Package for Constructing Synthetic Quasar Absorption Line Catalogs from Astrophysical Hydrodynamic Simulations". In: *The Journal of Open Source Software* 5.52, 2581 (Aug. 2020), p. 2581. DOI: 10.21105/joss.02581

Research Experience

Computational Nuclear Astrophysics

2022-Present

Type Ia Supernovae progenitor modeling using the MAESTROeX hydrodynamic code to study the Convective Urca Process during simmering phase of white dwarfs.

Galactic Modeling Research

2019-2020

Studied ENZO simulations produced by FOGGIE collaboration to better understand the Circumgalactic Medium (CGM). Generated synthetic spectra of the CGM to better inform/compare to observations.

MSU Campus Observatory

2018-2019

Assisted in data collection and reduction using 24-inch telescope. Observed cataclysmic variables, supernovae and transiting exoplanets.

HAWC Research Group

2016-2018

Studied gamma ray sources using the HAWC Observatory. Mainly worked on improving the detection sensitivity through machine learning techniques.

Computational Skills

Programming Experience

Proficient in Python

Proficient in C++

Competent in MPI parallelism

Competent in OpenMP threading

Competent in FORTRAN

Computational Fluid Modeling

Low Mach Flows

(Nuclear) Reactive Flows

Finite Volume Methods

Invited Talks

FSU Astronomy Seminar

2024

Presented work on hydrodynamic simulations of the Convective Urca Process at the Florida State University astronomy seminar.

SNEx Group

2024

Presented work on hydrodynamic simulations of the Convective Urca Process at the SNEx group; a group focused on type Ia SNe research, both observational and theoretical.

SBU IACS Student Seminar

2023

Presented work at the Stony Brook IACS student seminar on the challenges associated with using numerical fluid simulations to study astrophysics, with particular focus on Type Ia Supernovae.

Teaching Experience

Teaching Assistant - Stony Brook University

AST 248: “The Search for Life in the Universe”. A course designed to give an overview of the current knowledge of life outside of Earth and how we are searching for it. Topics such as habitability in our solar system, biosignatures, Fermi Paradox, etc.

Fall 2021, Spring 2021, Fall 2020

Undergraduate Learning Assistant - Michigan State University

AST 208: “Planets & Telescopes”, **AST 207:** “The Science of Astronomy”, **ISP 205:** “Visions of the Universe”.

Spring 2020, Fall 2019, Spring 2019

Conference Presentations

Nuclei in the Cosmos 2025

2025

Presented poster on new nuclear networks used in simulations of the Convective Urca Process at NIC - Nuclear Astrophysics conference.

American Astronomical Society Meeting 245

2025

Presented poster on recently published paper pertaining to the Convective Urca Process.

ASTRONUM 2024

2024

Presented a talk on simulations of the Convective Urca Process at ASTRONUM - International Conference on Numerical Modeling of Space Plasma Flows.

Nuclei in the Cosmos 2023

2023

Presented poster on simulations of the Convective Urca Process at NIC - Nuclear Astrophysics conference.

ASTRONUM 2023

2023

Presented a poster on simulations of the Convective Urca Process at ASTRONUM - International Conference on Numerical Modeling of Space Plasma Flows.

American Astronomical Society Meeting 241

2023

Presented poster on simulation of the Convective Urca Process at the winter AAS Meeting

Summer Schools Attended

International High Performance Computing Summer School

2022

Summer school for early-career computational scientists. Familiarized students with major state-of-the-art aspects of HPC and Big Data Analytics. Provided advanced mentoring and facilitated international networking.

Certification

CIMER Mentoring Training

2024

Completed a training workshop grounded in the Entering Mentoring series from the Center for the Improvement of Mentored Experiences in Research

Community Service and Outreach

MSU Observatory Public Nights

2018

Open house events at the MSU Observatory. Taught the public about astronomy and the work done at the observatory. Assisted people with looking through telescopes.

MSU Science Festival

2017-2018

Annual event used to inform and inspire the general public. Worked a station explaining the HAWC experiment and Cosmic Rays.

Tour de Ville

2016-2020

Annual charity bicycle ride put on by the Northville Rotary Club. Help with sending mass emails for the event as well as contributing the day of the ride (e.g. setting up of rest stops along course)

Honors and Awards

Peter B. Kahn Prize

2024

Awarded to Physics & Astronomy graduate student for outstanding research and travel.

Hantel Fellowship

2017, 2019

Awarded to Michigan State undergraduates conducting research in physics.

Michigan State Dean's List

2016-2020

Recognized for eight semesters as a student with at least a 3.5 GPA.